INCH-POUND

MIL-PRF-11693D
9 August 2001
SUPERSEDING
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8 October 1971

PERFORMANCE SPECIFICATION

CAPACITORS, FEED THROUGH, RADIO-INTERFERENCE REDUCTION,
DC (HERMETICALLY SEALED IN METAL CASES),
ESTABLISHED AND NON-ESTABLISHED RELIABILITY
GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for established reliability (ER) and non-ER capacitors designed for operation with direct current (dc), paper, paper plastic, and plastic (extended foil) dielectric, radio-interference-reduction, feed-through capacitors, hermetically sealed in metal cases (see 6.6.2), for use primarily in broadband, radio-interference suppression application. Capacitors meeting the established reliability requirements specified herein have a maximum failure rate of 1%/1000 hours. This failure rate is established with a 90% confidence limit based on the life test parameters specified and are maintained at a 10-percent producer's risk. An acceleration factor of 5:1 has been used to relate the life test data obtained at 140 percent of rated dc voltage at the applicable high test temperature to the rated voltage at the applicable high test temperature.

1.2 Classification.

1.2.1 Part or Identifying Number (PIN). The PIN is in the following form, and as specified (see 3.1 and 6.2):

| CZR23 | B | K | B | 473 | M |
|---|---|---|---|---|---|
| ER Style (1.2.1.1) | Current (1.2.1.2) | Characteristic (1.2.1.3) | Voltage (1.2.1.4) | Capacitance (1.2.1.5) | Failure rate level (1.2.1.6) |

| CZ23 | B | K | B | 473 |
|---|---|---|---|---|
| NON-ER Style 1.2.1.1 | Current 1.2.1.2 | Characteristic 1.2.1.3 | Voltage 1.2.1.4 | Capacitance 1.2.1.5 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Supply Center, Columbus, ATTN: DSCC/VAT, Post Office Box 3990, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5910

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1.2.1.1 Style. The style is identified by either the three-letter symbol “CZR” or the two-letter symbol “CZ” followed by a two-digit number. The letters identify dc, radio-interference reduction feed through capacitors, hermetically sealed in metal cases. The symbol “CZR” identifies established reliability (ER) capacitors; the symbol “CZ” identifies capacitors for which no specific reliability requirements are specified (non-ER). The first digit following the letter symbols identifies the general shape of the case, and the second digit identifies specific details other than case size. Each style designation may include a family of case sizes.

1.2.1.2 Current. The current rating is identified by a single letter in accordance with table I.

TABLE I. Current rating.

| Symbol | Current rating (dc amperes) |
|---|---|
| B | 10 |

1.2.1.3 Characteristic. The characteristic is identified by a single letter in accordance with table II.

TABLE II. Characteristic.

| | Values for characteristics | |
|---|---|---|
| | K | E |
| High ambient test temperature, degrees Celsius ±3°C | 125 | 85 |
| Low ambient test temperature, degrees Celsius ±3°C | -55 | -55 |
| Life-test voltage (see 4.7.19), in percent of rated dc voltage rating | 140 | 140 |
| Capacitance tolerance (percent) | ±10 | ±10 |

1.2.1.4 Voltage. The voltage rating is identified by a single letter in accordance with table III.

TABLE III. Voltage rating.

| Symbol | Voltage rating (Volts, dc) |
|---|---|
| B | 100 |
| C | 200 |
| E | 400 |
| F | 600 |

1.2.1.5 Capacitance. The nominal capacitance value expressed in picofarad (pF) is identified by a three-digit number; the first two digits represent significant figures and the last digit specifies the number of zeros to follow.

1.2.1.6 Failure rate level. The failure rate level of 1.0 percent per 1,000 hours is identified by a symbol M, and is based on operation at rated working voltage and maximum rated temperature.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.



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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-PRF-11693/7 - Capacitors, Feed Through, Radio-Interference Reduction, DC (Hermetically Sealed in Metal Cases), Established and Non-Established Reliability, Styles CZ23, CZ24, CZR23 and CZR24.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

DEPARTMENT OF DEFENSE

MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-220 - Method of Insertion Loss Measurement.
MIL-STD-690 - Failure Rate Sampling Plans and Procedures.
MIL-STD-790 - Established Reliability and High Reliability Qualified Products List (QPL) Systems for Electrical, Electronic, and Fiber Optic Parts Specifications.
MIL-STD-1285 - Marking of Electrical and Electronic Parts.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, Building 4D (DPM-DODSSP), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents that are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NCSL Z540-1-1994 - Calibration Laboratories and Measuring and Test Equipment, General Requirements for.

AMERICAN SOCIETY FOR TESTING AND MATERIALS

D92-98a - Standard Test Method for Flash and Fire Points by Cleveland Open Cup. (DoD adopted).

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA. 19428-2959.

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-554-1 - Assessment of Average Outgoing Quality Levels in Parts Per Million (ppm). (DoD adopted).
EIA-557 - Statistical Process Control Systems. (DoD adopted).



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(Application for copies should be addressed to the Electronic Industries Alliance (EIA), 2500 Wilson Boulevard, Arlington, VA 22201-3834.)

INTERNATIONAL ORGANIZATION FOR STANDARDS (ISO)

ISO 10012-1 - Quality Assurance Requirements for Measuring Equipment, Part 1: Meteorological Confirmation System for Measuring Equipment.

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheet, or MS sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern. (see 6.2)

3.2 Qualification. Capacitors furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) before contract award. In addition, the manufacturer shall obtain certification from the qualifying activity that the QPL system requirements of 3.3 and 4.4 have been met and are being maintained. Authorized distributors that are approved to MIL-STD-790 distributor requirements by the QPL manufacturers are listed in the QPL.

3.3 QPL system. The manufacturer shall establish and maintain an internal documentation system for qualified products covered by this specification. Requirements for this system are specified in MIL-STD-690 and MIL-STD-790. In addition, the manufacturer shall establish a Statistical Process Control (SPC) and Part Per Million (PPM) system that meets the requirements of 3.3.1 and 3.3.2, respectively. The following MIL-STD-790 exceptions are allowed:

(a) Under 'Description of production processes and controls', the procedure for identification of each production lot shall include only 'the manufacturer shall as a minimum be able to identify the time period during which the final production operation was performed on each item of product prior to final test. The date or lot code marked on each part shall be identified to a production lot.'

(b) 'Traceability' of materials shall not apply.

3.3.1 SPC system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a SPC system which meets the requirements of EIA-557. Typical manufacturing processes for application of a SPC include pre-assembly, assembly, encapsulation, and packaging.

3.3.2 PPM system. As part of the overall MIL-STD-790 QPL system, the manufacturer shall establish a PPM system for assessing the average outgoing quality of lots in accordance with EIA-554-1. Data exclusion, in accordance with EIA-554-1, may be used with approval of the qualifying activity. The PPM system shall identify the PPM rate at the end of each month and shall be based on a six month moving average.

3.4 Materials. The materials shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the capacitors to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.4.1 Impregnating and filling compounds. Compounds used in the impregnation and filling of capacitors shall be chemically inactive with respect to the capacitor element and the case (see 3.5.1). The compound either in the state of original application or as a result of having aged, shall have no adverse effect on the performance of the capacitor. For liquid-filled capacitors, the same material shall be used for impregnating as is used for filling.



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3.5 Interface and physical dimension requirements. Capacitors shall meet the interface and physical dimensions specified (see 3.1).

3.5.1 Case. Each capacitor shall be enclosed in a hermetically sealed metal case that will prevent leakage of the impregnant or filling compound, and will protect the capacitor element from moisture and mechanical damage under all test conditions specified herein.

3.5.2 Finish. All exposed metal surfaces shall be suitably protected against corrosion by plating, or other means (see 3.15). The resultant finish shall form a good electrical conductor and shall be free from defects that may affect its protective value.

3.5.3 Threaded parts. All threaded parts shall be as specified (see 3.1) and in accordance with FED-STD-H28. Aluminum nuts shall not be used.

3.5.3.1 Engagement of threaded parts. All threaded parts shall engage by at least two full threads.

3.6 Seal. When capacitors are tested as specified in 4.7.2, there shall be no continuous visible stream of bubbles or other evidence of leakage.

3.7 Terminal strength. When capacitors are tested as specified in 4.7.3, no part of the terminal shall loosen or rupture, and there shall be no other damage. Stud-type terminals shall exhibit no perceptible movement relative to the case, under the applied torque.

3.8 Dielectric withstanding voltage. When capacitors are tested as specified in 4.7.4, there shall be no visible damage, flashover, breakdown, open-circuiting or short-circuiting (see 6.6.3).

3.9 Insulation resistance. When measured as specified in 4.7.5, the insulation resistance, shall be not less than the applicable value specified in table IV and shown on figure 1. For measurements made at temperatures between 20° and 35°C, the applicable correction factor specified in table V shall be applied (see 6.6.3).

TABLE IV. Insulation resistance measurements.

| Capacitance rating | Minimum insulation resistance |
|---|---|
| Characteristic K | At 25°C |
| 0 to 0.33 microfarads | A 18,000 megohms |
| Greater than 0.33 microfarads | 6,000 megohm-microfarads 1/ |
| | At 125°C |
| 0 to 0.067 microfarads | 150 megohms |
| Greater than 0.067 microfarads | 10 megohm-microfarads 1/ |
| Characteristic E | At 25°C |
| 0 to 0.33 microfarads | 6,000 megohms |
| Greater than 0.33 microfarads | 2,000 megohm-microfarads 1/ |
| | At 85°C |
| 0 to 0.033 microfarads | 600 megohms |
| Greater than 0.033 microfarads | 20 megohm-microfarads 1/ |

1/ Product obtained by multiplying the capacitance in microfarads (µF) by the insulation resistance.



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TABLE V. Insulation-resistance correction factors.

| Temperature (Degrees C) | Correction factor 1/ | Temperature (Degrees C) | Correction factor 1/ | Temperature (Degrees C) | Correction factor 1/ | Temperature (Degrees C) | Correction factor 1/ |
|---|---|---|---|---|---|---|---|
| 20 | 1.42 | 24 | 1.08 | 28 | 0.82 | 32 | 0.63 |
| 21 | 1.33 | 25 | 1.00 | 29 | 0.76 | 33 | 0.59 |
| 22 | 1.24 | 26 | 0.94 | 30 | 0.71 | 34 | 0.55 |
| 23 | 1.16 | 27 | 0.87 | 31 | 0.67 | 35 | 0.51 |

1/ These correction factors are to be applied to the required values, not to the measured values.

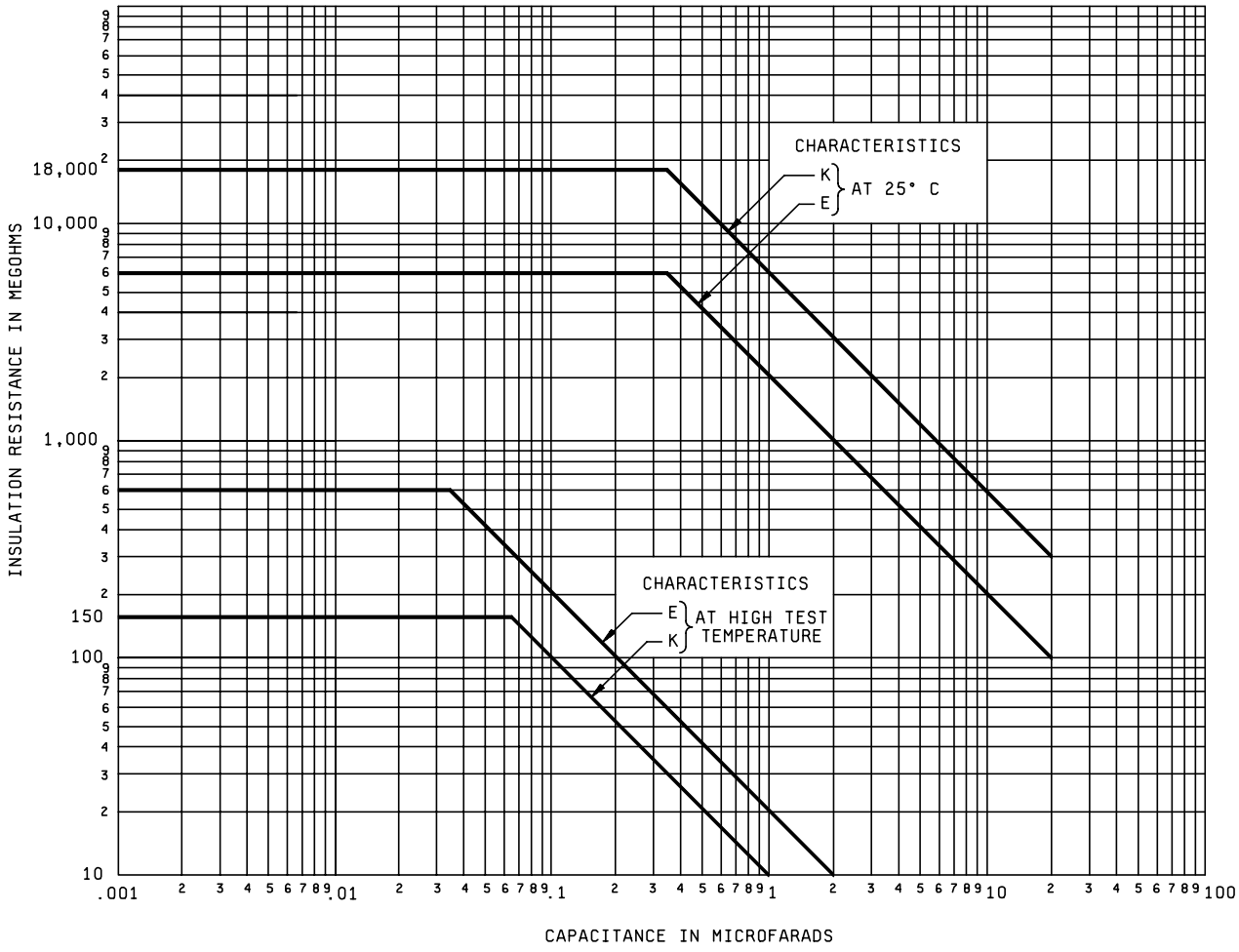



FIGURE 1. Insulation-resistance curve for characteristics K and E.



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3.10 Capacitance. When measured as specified in 4.7.6, the capacitance shall be not less than 90 percent nor greater than 110 percent of the specified nominal value (see 3.1).

3.11 Dissipation factor. When measured as specified in 4.7.7, the dissipation factor shall be not more than 1.0 percent.

3.12 Insertion loss. When measured as specified in 4.7.8, the insertion loss shall be not more than 6 decibels (dB) below the value shown on figure 2 (unless otherwise specified, see 3.1) for an ideal capacitor of the same nominal capacitance value. Dips are permissible when tested at frequencies from 0.15 megahertz (MHz) up to that frequency at which the insertion loss of the capacitor under test becomes 60 dB; however, at higher frequencies up to 1,000 MHz the insertion loss shall not fall below 60 dB. Deviations in the nature of dips in the curve will be permitted. Such dips indicate a drop in insertion loss below that of an ideal capacitor followed by a rise in insertion loss, as frequency is increased faster than the rise that is characteristic of an ideal capacitor. Such dips are also characteristic of all extended-foil capacitors.

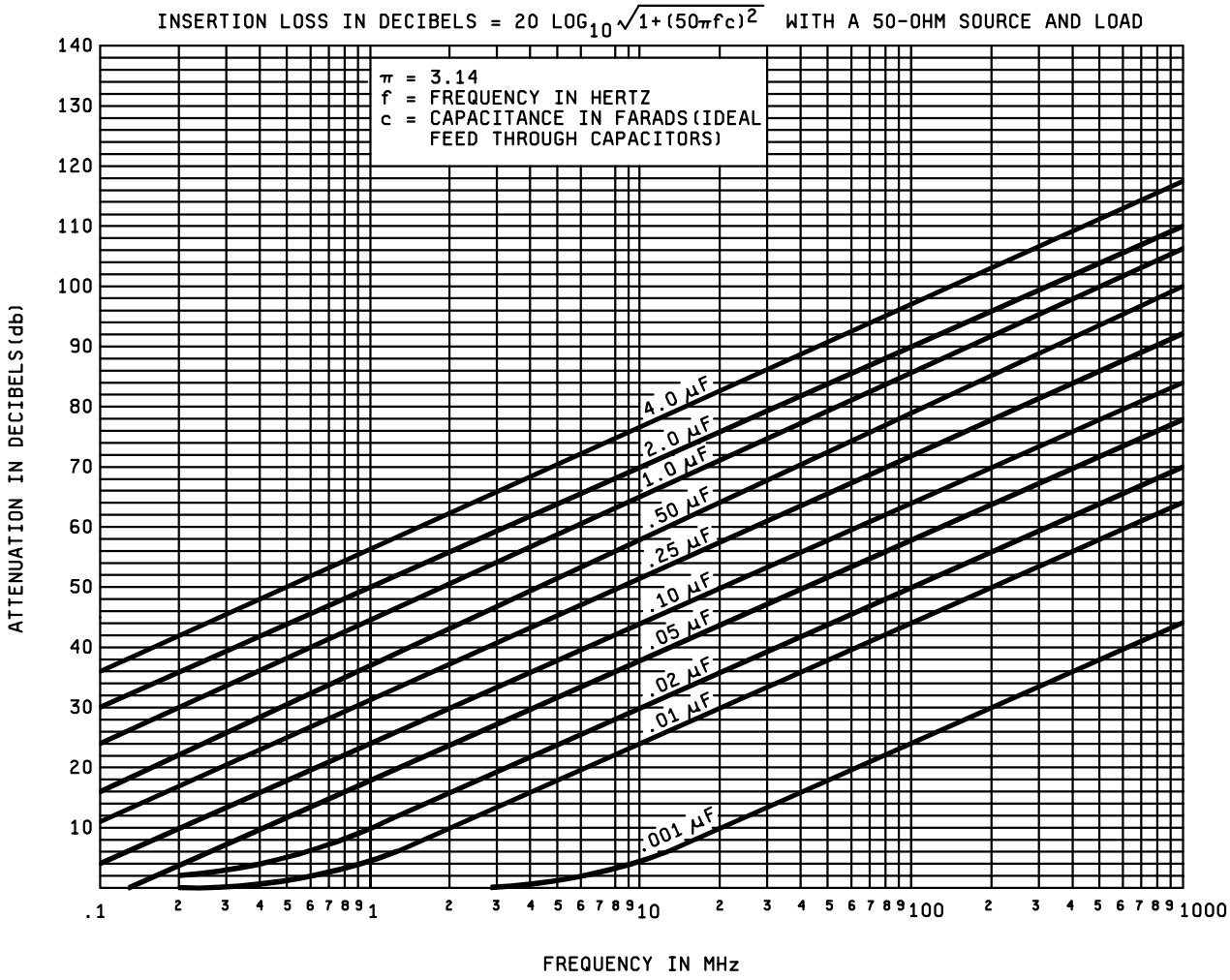



FIGURE 2. Insertion-loss versus frequency for an ideal capacitor.



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3.13 DC resistance. When measured as specified in 4.7.9, the dc resistance shall not exceed the value specified (see 3.1).

3.14 Resistance to soldering heat. When tested as specified in 4.7.10, there shall be no damage to the capacitors or the terminal insulators that will cause electrical failure of the capacitors, or which will cause hermetically sealed capacitors to leak. Chipping of terminal insulators alone shall not be cause for failure unless the chipping extends to the outer periphery. Capacitors shall also meet the following requirements:

| | |
|---|---|
| Insulation resistance | Not less than the value specified (see 3.1). |
| Capacitance | Change not more than the value specified (see 3.1). |
| Dissipation factor | Shall not exceed 1.5 percent. |

3.15 Salt spray (corrosion). When capacitors are tested as specified in 4.7.11, there shall be no harmful or extensive corrosion and at least 90 percent of any exposed metallic surfaces of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test.

3.16 Thermal shock and immersion. When tested as specified in 4.7.12, capacitors shall meet the following requirements:

| | |
|---|---|
| Dielectric withstanding voltage | As specified in 3.8. |
| Insulation resistance (at 25°C) | Not less than the value specified (see 3.1). |
| Capacitance | Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.6. |
| Dissipation factor | Shall not exceed 1.5 percent. |
| Insertion loss | Change at any frequency not more than 3 dB from the initial value obtained when measured as specified in 4.7.8 except that for those frequencies at which the initial value is 60 dB or greater, the change may exceed 3 dB provided the insertion loss does not drop below 60 dB. |
| Visual examination | There shall be no harmful or extensive corrosion and at least 90 percent of any exposed metal surfaces of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test. |

3.17 Shock (specified pulse). When capacitors are tested as specified in 4.7.13, there shall be no intermittent contacts of 0.5 millisecond (ms) or greater duration, open or short-circuiting and no mechanical damage to the capacitors. After completion of the shock test, the units shall be subjected to the insertion loss test (see 4.7.8). The change at any frequency shall be not more than 3 dB from the initial value obtained when measured as specified (see 3.1), except for those frequencies at which the initial value is 60 dB or greater, which may exceed 3 dB provided the insertion loss does not drop below 60 dB.

3.18 Vibration, high frequency. When capacitors are tested as specified in 4.7.14 there shall be no intermittent contacts of 0.5 ms or greater duration, open-circuiting, short-circuiting and no mechanical damage to the capacitors.

3.19 Moisture resistance. When tested as specified in 4.7.15, capacitors shall meet the following requirements:

| | |
|---|---|
| Dielectric withstanding voltage | As specified in 3.8. |
| Insulation resistance (at 25°C) | Not less than the value specified (see 3.1). |
| Capacitance | Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.6. |
| Dissipation factor | Shall not exceed 1.5 percent. |
| Visual examination | There shall be no harmful or extensive corrosion and at least 90 percent of any exposed metal surfaces of the capacitor shall be protected by the finish. In addition, there shall be not more than 10-percent corrosion of the terminal hardware or mounting surface. Marking shall remain legible after the test. |



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3.20 Solderability. When capacitors are tested as specified in 4.7.16, the dipped surface of the capacitor leads shall be at least 95 percent covered with a new continuous solder coating, free from pinholes or voids. The remaining 5 percent of the lead surface shall show only small pinholes or voids. These shall not be concentrated in one area. Bare base material and areas where the solder dip failed to cover the original coating are indications of poor solderability and shall be cause for failure. In case of dispute, the percentage covered by pinholes or voids shall be determined by actual measurement of those areas, as compared to the total area.

3.21 Resistance to solvents. When capacitors are tested as specified in 4.7.17, there shall be no evidence of mechanical damage, and the marking shall remain legible.

3.22 Low temperature and capacitance change at reduced temperature. When capacitors are tested as specified in 4.7.18, there shall be no evidence of breakdown, flashover, or fracture. The capacitance change at -55° ±3°C shall not exceed 15 percent of the value measured at 25°C.

3.23 Life. When tested as specified in 4.7.19, there shall be no mechanical failure or leakage of impregnant or filling compound and capacitors shall meet the following requirements:

| | |
|---|---|
| Insulation resistance (at 25°C)..... | Not less than the value specified (see 3.1). |
| Capacitance..... | Change not more than the value specified (see 3.1) from the initial value obtained when measured as specified in 4.7.6. |
| Dissipation factor ..... | Shall not exceed 1.5 percent. |
| Insertion loss..... | Unless otherwise specified (see 3.1), change at any frequency not more than 3 dB from the initial value obtained when measured as specified in 4.7.8, except that for those frequencies at which the initial value is 60 dB or greater, the change may exceed 3 dB provided the insertion loss does not drop below 60 dB. |

3.24 Flashpoint of impregnant or filling compound. When measured as specified in 4.7.20, the flashpoint of the impregnant or filling compound shall be not lower than 145°C for characteristic K, and 135°C for characteristics E.

3.25 Marking. Capacitors shall be marked with the type designation (Jan brand when applicable (see 3.25.2)) trade mark ™ and/or source code, date code, voltage and current ratings. The marking shall be placed on the exposed cylindrical surface of the case and shall remain legible after all environmental tests. Labels shall not be used. Other markings that in any way interfere with, obscure, or confuse those specified herein are prohibited. There shall be no space between the symbols that comprise the type designation. If space limitation requires it, the type designation may appear on two lines. In this event, the type designation shall be divided between the current symbol and the characteristic symbol as shown in the following examples:

| | | | |
|---|---|---|---|
| CZ23B | | CZR23B | |
| KB473 | | KB473M | |
| JAN ™ | | JAN ™ | |
| 12345 | or | 12345 | Source code |
| 7011 | | 7011 | Date code |
| 100 Volt | | 100 Volt | |
| 20 Amps | | 20 Amps | |

NOTE: The trademark may appear anyplace on the capacitor. Successful qualification is a prerequisite for “JAN” branding.

3.25.1 Date code. The date code shall be in accordance with MIL-STD-1285.

3.25.2 JAN and J marking. The United States Government has adopted, and is exercising legitimate control over, the certification marks “JAN” and “J” respectively, to indicate that items so marked or identified are manufactured to, and meet all the requirements of specifications. Accordingly, items acquired to and meeting all of the criteria specified herein and in applicable specification, shall bear the certification mark ”JAN”, except that items too small to bear the certification mark “JAN” shall bear the letter “J”. The “JAN” or “J” shall be placed immediately before the part number except that if such location would place a hardship on the manufacturer in connection with such marking, the "JAN“ or "J" may be located on the first line above or below the part number. Items furnished under



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contracts or orders which either permit or require deviation from the conditions or requirements specified herein and in applicable specifications shall not bear “JAN” or “J”. In the event an item fails to meet the requirements of this specification and the applicable specification sheets or associated specifications, the manufacturer shall remove completely the military part number and the "JAN" or the “J” from the sample tested and also from all items represented by the sample. The “JAN” or “J” certification mark shall not be used on products acquired to contractor drawings or specifications. The United States Government has obtained Certificate of Registration Number 504,860 for the certification mark “JAN” and Registration Number 1,586,261 for the certification mark “J”.

3.26 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.27 Workmanship. Capacitors shall be processed in such a manner as to be uniform in quality and shall be free from pits, corrosion, cracks, rough edges, and other defects that will affect life, serviceability, or appearance.

3.27.1 Soldering. Flux for soldering shall be rosin, rosin and alcohol, or rosin and turpentine. No acid or acid salts shall be used in preparation for or during soldering. All excess flux and solder shall be removed. Electrical connections shall be mechanically secured before soldering, where possible, and electrically continuous after soldering.

3.27.2 Terminal. Solder lugs and solder lug terminals may be of any shape provided dimensional limits are met. All terminals shall be coated with solder having a tin content of 40 to 70 percent.

## 4. VERIFICATION

4.1 Classification of inspections. The inspections specified herein are classified as follows:

(a) Qualification inspection (see 4.4).

(b) Verification of qualification (ER styles - see 4.5); (non-ER - see 4.5.1).

(c) Conformance inspection (see 4.6).

4.2 Test equipment and inspection facilities. The supplier shall establish and maintain a calibration system in accordance with ANSI/NCSL Z540-1, ISO10012-1, or equivalent, as approved by the qualifying activity.

4.3 Inspection conditions. Unless otherwise specified herein, all inspections shall be performed in accordance with the test conditions specified in the "GENERAL REQUIREMENTS" of MIL-STD-202 except relative humidity shall not exceed 75 percent. Accuracy of all test voltage measurements shall be within ±2.0 percent of the specified voltage.

4.4 Qualification inspection. Qualification inspection shall be performed at a laboratory acceptable to the Government (see 6.3) on sample units produced with equipment and procedures normally used in production.

4.4.1 Sample size. The number of capacitors to be subjected to qualification inspection shall be as specified in table VI and the appendix to this specification.

4.4.2 Inspection routine. The sample shall be subjected to the inspections specified in table VI, in the order shown. Two specimens will be subjected to the subsequent examinations and tests in group 1. The specimens will then be divided into the remaining groups specified in table VI, and subjected to the tests for their particular group.

4.4.3 Failures. Failures in excess of those allowed in table VI shall be cause for refusal to grant qualification approval.



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TABLE VI. Qualification Inspection.

| Inspection | Requirement paragraph | Test method paragraph | NON-ER | | ER | |
|---|---|---|---|---|---|---|
| | | | Number of sample units to be inspected | Number of defects permitted 1/ | Number of sample units to be inspected | Number of defects permitted |
| Group I | | | | | | |
| Visual and mechanical examination (internal): | | | | | | |
| Material and workmanship | 3.1, 3.4 to 3.5.3.1, and 3.25 to 3.27.2 | 4.7.1 | 2 | 0 | 2 | 0 |
| Visual and mechanical examination (external): | | | | | | |
| Physical dimensions, marking, and workmanship 2/ | 3.1, 3.5, and 3.25 and 3.27 | 4.7.1 | | | | |
| Seal | 3.6 | 4.7.2 | | | | |
| Terminal strength | 3.7 | 4.7.3 | | | | |
| Dielectric withstanding voltage | 3.8 | 4.7.4 | 29 3/ | 1 | 57 3/ | 1 |
| Insulation resistance | 3.9 | 4.7.5 | | | | |
| Capacitance | 3.10 | 4.7.6 | | | | |
| Dissipation factor | 3.11 | 4.7.7 | | | | |
| Insertion loss | 3.12 | 4.7.8 | | | | |
| DC resistance | 3.13 | 4.7.9 | | | | |
| Group II | | | | | | |
| Resistance to soldering heat (solderable terminals only) | 3.14 | 4.7.10 | | | | |
| Salt spray (corrosion) | 3.15 | 4.7.11 | 6 | 1 | 6 | 1 |
| Thermal shock and immersion | 3.16 | 4.7.12 | | | | |
| Group III | | | | | | |
| Shock (specified pulse) | 3.17 | 4.7.13 | 6 | | 6 | |
| Vibration, high frequency | 3.18 | 4.7.14 | | | | |
| Moisture resistance | 3.19 | 4.7.15 | | | | |
| Group IV | | | | | | |
| Solderability | 3.20 | 4.7.16 | 4 | | 4 | |
| Resistance to solvents 5/ | 3.21 | 4.7.17 | | | | |
| Group V | | | | | | |
| Low temperature and capacitance change at reduced temperature | 3.22 | 4.7.18 | 12 | 1 | 40 | 1 |
| Life (accelerated) 4/ | 3.23 | 4.7.19 | | | | |

1/ A specimen having one or more defects will be considered as a single defective.
2/ Marking will be considered as a defect only if it becomes illegible as a result of any of the tests of the specification.
3/ One additional specimen is included in each sample of 31 or 59 specimens as applicable to permit substitution for the allowable defective in group I.
4/ The decision as to whether or not the NON-ER product is to be included on the qualified products list shall be made at the end of the 240 hour life test and at the conclusion of the 2,000-hour life test for the ER parts.
5/ Two capacitors with sleeves and two unsleeved.



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4.4.4 Failure rate (FR) qualification (applicable to ER parts only). FR qualification for capacitors shall be in accordance with the general and detailed requirements of MIL-STD-690 and as follows.

a. Procedure I: Qualification at the initial FRL. Level "M" (1.0 percent) of FRSP-90 shall apply. Sample units shall be subjected to the qualification inspection specified in group V, table VI (see 4.4.2).

b. Procedure II: Extension of qualification to lower FRLs. To extend qualification to the "R" (0.01 percent) and "S" (0.001 percent) FRLs, data from two or more styles from the same specification sheet may be combined.

c. Procedure III: Maintenance of FRL qualification. Maintenance period B of FRSP-10 shall apply. Regardless of the number of production lots produced during this period, the specified number of unit hours shall be accumulated to maintain qualification.

4.4.5 Quality level verification (ER styles only). The manufacturer is responsible for establishing a quality system to assess the ppm defect level of lots that are subjected to subgroup 2 tests of the group A inspections. The ppm defect level shall be based on a 6-month moving average.

4.5 Verification of qualification (ER styles). Every 6 months the manufacturer shall compile a summary of the results of qualification conformance inspections and extended FR test data, in the form of a verification of qualification report, and forward it to the qualifying activity whenever the FR data indicates that the manufacturer has failed to maintain the qualified FR level, or the group B inspection data indicates failure of the qualified product to meet the requirements of this specification. Continuation shall be based on evidence that over the 6-month period the following has been met:

(a) MIL-STD-790 program.

(b) The capacitor design has not been modified.

(c) Lot rejection for group A inspection does not exceed 5 percent or one lot, whichever is greater.

(d) The requirements for group B inspection are met.

(e) Verification of FRLs.

(f) PPM assessment.

When group B requirements were not met and the manufacturer has taken corrective action satisfactory to the qualifying activity, group B retesting shall be instituted. A summary of the retesting shall be forwarded to the qualifying activity within 30 days after completion of the retest.

In the event that there is no production of a single style device during a maintenance period and the manufacturer is listed for more than one style on the QPL, the manufacturer shall certify that they retain the capabilities and facilities necessary to produce that product. However, the manufacturer shall still maintain the required number of unit hours in the maintenance period using those styles produced in order to remain qualified to the applicable failure rate levels. Styles need not be manufactured for testing only but the manufacturer must certify that the capability and facilities needed to produce that style are still in place. In the event that units must be built for the purpose of maintaining the required hours, they shall also undergo all required testing prior to being placed on life test and be subjected to all group B requirements.



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4.5.1 Verification of qualification (non-ER styles). To retain qualification, the supplier shall forward a report at 6-month intervals to the qualifying activity. The qualifying activity shall establish the initial reporting date. The report shall consist of:

(a) A summary of the results of the tests performed for inspection of product for delivery (group A), indicating as a minimum the number of lots that have passed and the number that have failed. The results of tests of all reworked lots shall be identified and accounted for.

(b) A summary of the results of tests performed for periodic inspection (group B), including the number and mode of failures. The summary shall include results of all periodic inspection tests performed and completed during the 6-month period. If the summary of the test results indicates nonconformance with specification requirements, and corrective action acceptable to the qualifying activity has not been taken, action may be taken to remove the failing product from the qualified products list.

Failure to submit the report within 30 days after the end of each 6 month period may result in loss of qualification for the product. In addition to the periodic report, the supplier shall immediately notify the qualifying activity at any time during the 6 month period that the inspection data indicates failure of the qualified product to meet the requirements of this specification.

In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during 3 consecutive reporting periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit (the products, a representative product of each type, grade, class, etc.) to testing in accordance with the qualification inspection requirements.

4.6 Conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of group A inspection. Except as specified in 4.6.2.3, delivery of products which have passed group A inspection shall not be delayed pending the results of group B inspection.

4.6.1.1 Inspection lot. An inspection lot shall consist of all capacitors of the same style, current and voltage ratings produced under essentially the same conditions, and offered for inspection at the same time. ER parts shall be kept separate from non-ER parts.

4.6.1.2 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table VII in the order shown.

4.6.1.2 1 Sampling plan. The sampling plan shall be as specified in table VII. In the event of one or more failures, the lot shall be rejected.

4.6.1.2.2 Manufacturers production inspection for ER parts only. If the manufacturer performs tests similar to those specified in subgroup 1, table VII, as the final step of his production process, group A, subgroup 1 inspection may be waived and the data resulting from the manufacturer's production tests may be used instead. Authority to waive the subgroup 1 inspection shall be granted by the qualifying activity only. The following criteria must be complied with:

(a) Tests conducted by the manufacturer during production shall be clearly identical to or more stringent than that specified for subgroup 1.

(b) Manufacturer subjects 100% of the product supplied under this specification to his production tests.

(c) The parameters measured and the failure criteria shall be the same or more stringent than those specified herein.

(d) The lot rejection criteria is the same or more stringent than that specified herein.



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TABLE VII Group A inspection.

| Examination or test | Requirement paragraph | Method paragraph | Sampling procedure | |
|---|---|---|---|---|
| | | | NON-ER | ER |
| Subgroup 1 | | | | |
| Seal | 3.6 | 4.7.2 | | |
| Dielectric withstanding voltage | 3.8 | 4.7.4 | | Not applicable. |
| Insulation resistance | 3.9 | 4.7.5 | See table VIII | (100% inspection) |
| Capacitance | 3.10 | 4.7.6 | | |
| Dissipation factor | 3.11 | 4.7.7 | | |
| Insertion loss | 3.12 | 4.7.8 | | |
| DC resistance | 3.13 | 4.7.9 | | |
| Subgroup 2 | | | | |
| Visual and mechanical examination (external): | | | | |
| Materials | 3.4 and 3.4.1 | 4.7.1 | 13 samples | See table VIII |
| Physical dimensions | 3.5 | | 0 failures | |
| Design construction (other than physical dimensions) | 3.5 to 3.5.3.1 | | | |
| Marking 1/ | 3.25 | | | |
| Workmanship | 3.28 | | | |

1/ Marking defects are based on visual examination only and shall be charged only for illegible, incomplete, or incorrect markings. Any subsequent electrical defects shall not be used as a basis for determining marking defects.

TABLE VIII. Sampling plans for Group A tests.

| Lot size | | Sample size |
|---|---|---|
| 1- | 13 | 100% |
| 14- | 150 | 13 |
| 151- | 280 | 20 |
| 281- | 500 | 29 |
| 501- | 1,200 | 34 |
| 1,201- | 3,200 | 42 |
| 3,201- | 10,000 | 50 |
| 10,001- | 35,000 | 60 |
| 35,001- | 150,000 | 74 |
| 150,001- | 500,000 | 90 |
| 500,001- | Up | 102 |

4.6.1.2.3 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots. If during the 100 percent inspection of subgroup 1 (ER parts only), screening requires that over 5 percent of the capacitors be discarded, the lot shall be rejected and shall not be supplied to this specification.

4.6.1.2.4 PPM calculations (ER and non-ER). The manufacturer shall establish and maintain a system for assessing and calculating average outgoing quality of capacitors. A PPM rate combining IR, capacitance, and DF shall be assessed for lots that have passed the group A inspection. The manufacturer's PPM system shall also address rectification procedures for lots failing PPM assessment. Data from the rectification process shall not be used to calculate PPM.



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4.6.2 Periodic group B inspection. Periodic group B inspection shall consist of the tests specified in table IX, in the order shown, and shall be performed on sample units selected from lots that have passed group A inspection. Except where the results of these inspections show noncompliance with the applicable requirements (see 4.6.2.3), delivery of products that have passed group A shall not be delayed pending the results of this periodic inspection.

TABLE IX. Group B inspection.

| Inspection | Requirement paragraph | Method paragraph | Number of sample units to be inspected | Number of defects permitted | |
|---|---|---|---|---|---|
| | | | | NON-ER | ER |
| Subgroup 1 | | | | | |
| Terminal strength 1/ | 3.7 | 4.7.3 | 6 | 1 | 1 |
| Insertion loss | 3.12 | 4.7.8 | | | |
| Subgroup 2 | | | | | |
| Resistance to soldering heat (soldered terminals only) 1/ | 3.14 | 4.7.10 | 6 | 1 | 1 |
| Salt spray (corrosion) 1/ | 3.15 | 4.7.11 | | | |
| Thermal shock and immersion 1/ | 3.16 | 4.7.12 | | | |
| Subgroup 3 | | | | | |
| Shock (specified pulse) 2/ | 3.17 | 4.7.13 | 6 | 1 | 1 |
| Vibration (high frequency) 2/ | 3.18 | 4.7.14 | | | |
| Moisture resistance 1/ | 3.19 | 4.7.15 | | | |
| Subgroup 4 3/ | | | 3 | 1 | 1 |
| Solderability | 3.20 | 4.7.16 | | | |
| Resistance to solvents 1/ | 3.21 | 4.7.17 | | | |
| Subgroup 5 (NON-ER parts only) | | | 6 | 1 | N/A |
| Low temperature and capacitance change at reduced temperature | 3.22 | 4.7.18 | | | |
| Insulation resistance (at the high ambient temperature) | 3.9 | 4.7.5 | | | |
| Life (240 hours) | 3.23 | 4.7.19.2.1 | | | |
| Subgroup 6 (ER parts only) | | | 6 | N/A | 1 4/ |
| Low temperature and capacitance change at reduced temperature | 3.22 | 4.7.18 | | | |
| Insulation resistance (at the high ambient temperature) | 3.9 | 4.7.5 | | | |
| Life (2,000 hours) | 3.23 | 4.7.19.2.2 | | | |

1/ If the manufacturer can demonstrate that this test has been performed five consecutive times with zero failure, this test, with the approval of the qualifying activity, can be deleted. The manufacturer, however, shall perform this test every 3 years after the deletion as part of long term design verification. If the design, material, construction processing of the part is changed, or if there are any quality problems, the qualifying activity may require resumption of the specified testing. Deletion of testing does not relieve the manufacturer from meeting the test requirements in case of dispute.
2/ The time requirement for shock and vibration tests is a minimum of once every 24 months.
3/ Two capacitors with sleeve and one unsleeved.
4/ The number of defects is also governed by the requirement of 4.5, wherein the qualified product must meet the 90 percent failure rate plan on an annual basis.



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4.6.2.1 Sampling plan. Sample units shall be representative of capacitors produced.

4.6.2.1.1 Subgroups 1, 2, 3, and 4 (ER and non-ER). Six sample units shall be taken from production every 2 months and subjected to the applicable tests for their particular subgroup. Allowable failures shall be as specified in table IX.

4.6.2.1.2 Subgroup 5 (non-ER parts). Six sample units shall be taken from production every 2 months and subjected to the applicable tests for their particular subgroup. Allowable failures shall be as specified in table IX.

4.6.2.1.3 Subgroup 6 (ER parts only). Six sample units shall be selected from lots produced during a 6-month period. Allowable failures shall be as specified in table IX. The accumulated life test data shall be used for maintenance of FR qualification, and shall be verified for failure rate level maintenance with 90 percent confidence level, at 2-year intervals.

4.6.2.2 Disposition of sample units. Sample units which have been subjected to group B inspection shall not be delivered on the contract or order.

4.6.2.3 Noncompliance (applicable to both ER and non-ER parts). If a sample fails to pass group B inspection, the supplier shall take corrective action on the materials or processes, or both, as warranted, and on all units of product which can be corrected and which were manufactured under essentially the same conditions, with essentially the same materials, processes, etc., and which are considered subject to the same failure. Acceptance of the product shall be discontinued until corrective action, acceptable to the Government, has been taken. After the corrective action has been taken, group B inspection shall be repeated on additional sample units (all inspection, or the inspection which the original sample failed, at the option of the Government). Group A inspection may be re-instituted; however, final acceptance shall be withheld until group B re-inspection has shown that the corrective action was successful. In the event of failure after reinspection, information concerning the failure and corrective action taken shall be furnished to the cognizant inspection activity and the qualifying activity.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. Capacitors shall be examined to verify that the materials, design, construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements (see 3.1, 3.4 to 3.5.3.1, 3.25 and 3.27).

4.7.2 Seal (see 3.6).

4.7.2.1 Solid-filled, liquid-impregnated, and characteristic P capacitors. While at room temperature, capacitors whose filling material will be solid at the high ambient test temperature, and liquid-impregnated capacitors in which no filling compound is used, shall be immersed for at least 1 minute in oil or water maintained at a temperature of 5°C +5°C, -0°C above the high ambient test temperature.

4.7.2.2 Liquid-filled capacitors. Capacitors whose filling material will be liquid at the high ambient test temperature, shall be placed with the terminals facing horizontal and brought to a case temperature of 5°C, +5°C, -0°C above the high ambient test temperature. After 10 minutes at this case temperature, the capacitors shall be turned through 180° onto another surface with the terminals still horizontal and kept in this position for 10 minutes with the case temperature maintained within the same limits. An oil spot or spots appearing on the paper or any oil impregnant appearing on the outside of the case is considered evidence of leakage.



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4.7.3 Terminal strength (see 3.7). Capacitors shall be tested in accordance with method 211 of MIL-STD-202. The following details and exceptions shall apply:

(a) Pull test - applies to all terminals.

1. Test condition letter: A.
2. Method of holding: normal mounting means.
3. Applied force: 5 pounds +3, -0, ounces.
4. Duration of applied force: 30 +5, -0 seconds.

(b) Torque - threaded or tapped stud type terminals only.

1. Test condition letter: E.
2. Torque applied to nuts or screws shall be as specified in table X.
3. Applications: A total of five tightenings shall be applied.

TABLE X. Torque.

| Screw size | Torque (inch-pounds) |
|---|---|
| 8-32 | 14 |
| 10-32 | 18 |
| 1/4-28 | 24 |
| 5/16-24 | 50 |
| 3/8-24 | 100 |
| 7/16-20 | 150 |
| 1/2-28 | 200 |

4.7.4 Dielectric withstanding voltage (see 3.8).

4.7.4.1 For qualification inspection. Capacitors shall be tested in accordance with method 301 of MIL-STD-202. The following details shall apply:

(a) Magnitude and nature of test voltage: 200 percent of the rated dc voltage. The ac ripple shall not exceed 2 percent.

(b) Duration of application of test voltage: 1 minute ±5 seconds.

(c) Points of application of test voltage: Between the case and either terminal.

(d) Limiting value of surge current: Between 5 milliamperes and 1 ampere.

(e) Measurement during and after dielectric withstanding voltage: after the test, capacitors shall be visually examined for external evidence of damage, arcing, or breakdown.

4.7.4.2 For acceptance inspection. Capacitors shall be tested as specified in 4.7.4.1, except that, at the option of the manufacturer, a test voltage of 250 percent of the rated dc voltage may be applied for a period of not less than 1 second.



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4.7.5 Insulation resistance (see 3.9). Capacitors shall be tested in accordance with method 302 of MIL-STD-202. The following details and exceptions shall apply:

(a) Test potential: A potential equal to the rated voltage (see 3.1) or 500 volts dc, whichever is less.

(b) Special conditions: Capacitors shall be tested at the applicable high ambient test temperature (see table II), and at 25°C or corrected thereto.

(c) Points of measurement: Between either terminal and the case.

4.7.6 Capacitance (see 3.10). Capacitors shall be tested in accordance with method 305 of MIL-STD-202. The following details shall apply:

(a) Test frequency: 1 kHz ±100 Hz for capacitors whose nominal capacitance is 1 µF or less, and 60 ±6 Hz for capacitors whose nominal capacitance is more than 1 µF.

(b) Limit of accuracy: Within ±2 percent.

4.7.7 Dissipation factor (see 3.11). The dissipation factor of each capacitor shall be measured at a root-mean-square (rms) voltage not greater than 20 percent of the rated dc voltage. Measurement shall be made at a frequency of 1 kHz ±100 Hz for capacitors whose nominal capacitance is more than 1 µF. Measurement accuracy shall be within ±0.1 percent absolute.

4.7.8 Insertion loss (see 3.12). Capacitors shall be tested in accordance with MIL-STD-220. Measurements shall be taken at the frequencies specified (see 3.1).

4.7.9 DC resistance (see 3.13). The dc resistance shall be measured directly using a Kelvin bridge or indirectly using the voltmeter-ammeter method, ammeter-potentiometer method, or other suitable means. The point of contact shall be along the lead, where the lead is normally connected in an actual circuit, except that for axial-wire leads, the contact shall be made 5/8 ±1/8 inch from the insulator. The maximum allowable measurement error shall be 5 percent.

4.7.10 Resistance to soldering heat, solderable terminals only (see 3.14).

(a) Depth of immersion in molten solder: Within .062 ±.031 inch of the seal or case.

(b) Test condition: B.

(c) Cooling time prior to measurement after test: 1 hour.

(d) Measurement after test: capacitance to ground and coil continuity.

4.7.11 Salt spray (corrosion) (see 3.15). Capacitors shall be tested in accordance with method 101, test condition B, of MIL-STD-202. The salt solution concentration shall be 5 percent. After this test, capacitors shall be visually examined for corrosion and obliteration of marking.

4.7.12 Thermal shock and immersion (see 3.16).

4.7.12.1 Thermal shock. Capacitors shall be tested in accordance with method 107 of MIL-STD-202. The following details shall apply:

(a) Test condition letter: A, except that in step 3, capacitors shall be tested at the applicable high ambient test temperature (see table II).

(b) Measurements before and after cycling: Not applicable.



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4.7.12.2 Immersion. Within 4 to 24 hours after completion of thermal shock, capacitors shall be tested in accordance with method 104 of MIL-STD-202. The following details shall apply:

(a) Test condition letter: B.

(b) Measurements after final cycle: Dielectric withstanding voltage, insulation resistance at 25°C, capacitance, dissipation factor, and insertion loss shall be measured as specified in 4.7.4, 4.7.5, 4.7.6, 4.7.7 and 4.7.8, respectively. These measurements shall be made within 2 hours after capacitors are removed from the immersion tank, except that no time limit is specified for the insertion-loss measurement. If the capacitors fail the insulation-resistance test, they may be retested for insulation resistance after a total maximum drying period of 24 hours.

(c) Visual examination: Capacitors shall be examined for corrosion and obliteration of marking.

4.7.13 Shock, (specified pulse) (see 3.17). Capacitors shall be tested in accordance with method 213 of MIL-STD-202. The following details shall apply:

(a) Mounting: securely fastened by normal mounting means. When applicable, leads and connecting wires must be supported to prevent damage to the capacitor.

(b) Test condition: Non ER - G (50g, peak);
ER - I (100g, peak).

(c) Electrical loading during shock: During the test, a potential of 125 percent of dc rated voltage (see 3.1) shall be applied to the capacitor.

(d) Measurements during shock: During the test, observations shall be made to determine intermittent contact or arcing or open-circuiting or short-circuiting. Detecting equipment shall be sufficiently sensitive to detect any interruption with a duration of 0.5 ms duration or greater.

(e) Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.

4.7.14 Vibration, high frequency (ER only) (see 3.18). Capacitors shall be tested in accordance with method 204 of MIL-STD-202. The following details shall apply:

(a) Mounting: securely fastened by normal mounting means. When applicable, leads and connecting wires must be so supported to exclude spurious resonance.

(b) Electrical load conditions: during the test, a potential of 125 percent of rated voltage (see 3.1) shall be applied between the terminal and case of the capacitor.

(c) Test condition letter: B (15g).

(d) Direction and duration of motion: 4 hours in each of two mutually perpendicular directions (total of 8 hours), one parallel and the other perpendicular to the cylindrical axis.

(e) Measurements during vibration: during the last half hour of vibration in each direction, a signal of 1 ±0.2 kHz at a level of 1 ± 0.2 volts shall be placed across the capacitor and measured with a suitable device to determine open-circuiting or short-circuiting or intermittent of 0.5 ms or greater duration.

(f) Examination after test: Capacitors shall be visually examined for evidence of mechanical damage.



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4.7.15 Moisture resistance (see 3.19). Capacitors shall be tested in accordance with method 106 of MIL-STD-202. The following details and exceptions shall apply:

(a) Initial measurements: Not applicable.

(b) Polarization voltage: The dc potential shall be applied to either terminal and the case of 50 percent of the capacitors. No potential shall be applied to the remaining 50 percent of the capacitors.

(c) Final measurements: Upon completion of step 6 of the final cycle, the dielectric withstanding voltage, insulation resistance at 25°C, capacitance, and dissipation factor shall be measured as specified in 4.7.4, 4.7.5, 4.7.6 and 4.7.7, respectively, at a relative humidity of 50 ±5 percent.

(d) Visual examination: Capacitors shall be examined for corrosion and obliteration of marking.

4.7.16 Solderability (wire leads only) (see 3.20). Capacitors shall be tested in accordance with method 208 of MIL-STD-202. The following details shall apply:

(a) The number of terminations of each capacitor to be tested: 2.

(b) Special preparation of the terminations: Prior to the application of the flux and solder dip, 50 percent of the capacitors shall be subjected to aging by immersion in a noncorrosive container of distilled water. For characteristic K capacitors, the distilled water shall be boiling. For characteristic E capacitors, the distilled water shall be maintained at 85°C. The immersion period shall be one hour. No aging shall be required on the remaining 50 percent of the capacitors.

(c) Depth of immersion in flux and solder: Both leads shall be immersed to within 0.05 inches from the point of egress.

4.7.17 Resistance to solvents (see 3.21). Capacitors shall be tested in accordance with method 215 of MIL-STD-202. The following details shall apply:

(a) Portion of specimen to be brushed: That portion on which marking is present.

(b) Number of specimens to be tested: As specified in applicable inspection tables.

(c) Permissible extent of damage: As specified in 3.21.

4.7.18 Low temperature and capacitance change at reduced temperature (see 3.22).

4.7.18.1 Low temperature. Capacitors shall be placed in a chamber maintained at -55° ±3°C, and rated voltage at the rated frequency (when applicable) shall be applied at this condition for 48 ±4 hours (see 3.1). The air within the conditioning chamber shall be circulated.

4.7.18.2 Capacitance change at reduced temperature. At the conclusion of the test specified in 4.7.18.1 but before the capacitors are removed from the conditioning chamber, the capacitance shall be measured as specified in 4.7.6 except that the measurement shall be made at -55° ±3°C. This measurement shall be compared with a measurement made at room ambient temperature. After exposure, capacitors shall be visually examined for evidence of breakdown, flashover, or fractures.



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4.7.19 Life (see 3.23).

4.7.19.1 Qualification inspection. Capacitors shall be tested in accordance with method 108 of MIL-STD-202. The following details and exceptions shall apply:

(a) Method of mounting and distance between specimens: Suspended by their terminals with a distance of not less than 1.5 inches separation between units.

(b) Test temperature and tolerance: The applicable high temperature specified in table II (see 3.1).

(c) Operating conditions: A potential of 140 percent of the rated dc voltage shall be applied between the terminal and the case. Rated current shall be flowing through the feed through conductors (see 3.1).

(d) Test condition: F (2,000 +96, -0 hours) for ER parts and 240 +48, -0 for non-ER parts.

(e) Measurements after exposure: Capacitors shall be returned to the inspection conditions specified in 4.3, and shall be visually examined for leakage of impregnant or filling compound and deformation of case. Insulation resistance at 25°C, capacitance, dissipation factor, and insertion loss shall then be measured as specified in 4.7.5, 4.7.6, 4.7.7 and 4.7.8, respectively, (see 3.23).

4.7.19.2 Life (Conformance inspection).

4.7.19.2.1 240 hour (group B inspection) (see 3.23). Capacitors shall be tested as specified in 4.7.19.1, except that the duration of the test shall be 240 +48, 0 hours.

4.7.19.2.2 2,000 hour (group B inspection) (ER parts only) (see 3.23). Capacitors shall be tested as specified in 4.7.19.1, except the duration of the test shall be 2,000 +96, -0 hours.

4.7.20 Flashpoint of impregnant or filling compound (see 3.24). This test is applicable only to liquid impregnants and filling compounds and to solid impregnants having a melting point of less than 135°C. The flashpoint of the impregnant or filling compound shall be measured as specified in ASTM D92-98a, except that fire point and precision do not apply. “Impregnant or filling compound” shall be substituted for the word “oil” throughout the test method.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The capacitors covered by this specification are primarily designed for use in broadband, radio-interference suppression applications. They are military unique due to the fact that they must be able to operate satisfactorily in military systems under the following demanding conditions: -55°C to +85°C or +125°C operating temperature range, 15 Gs of vibration, 50 - 100 Gs of shock, 48 hours of salt spray, 2,000 hours of use at the maximum temperature and 140% of rated voltage. In addition these military requirements are verified under a qualification system. Commercial components are not designed to withstand these military environmental conditions.



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6.2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of the specification.

(b) Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

(c) Packaging requirements (see section 5.1).

(d) Direct Governmental purchases should specify machine nuts and washers.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, prior to the award of contract, qualified for inclusion in the applicable qualified products list, whether or not such products have actually been so listed by that date. The attention of the contractors is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the qualified products list is the Defense Supply Center Columbus, ATTN: DSCC/VQP, PO Box 3990, Columbus, OH 43216-5000.

6.4 Insertion loss for ideal capacitors. The insertion loss of ideal capacitors for capacitance values from 0.001 to 100 µF, inclusive, and frequency range from 0.01 to 1,000 megahertz, inclusive, may be determined from figure 3 (see 4.7.8).

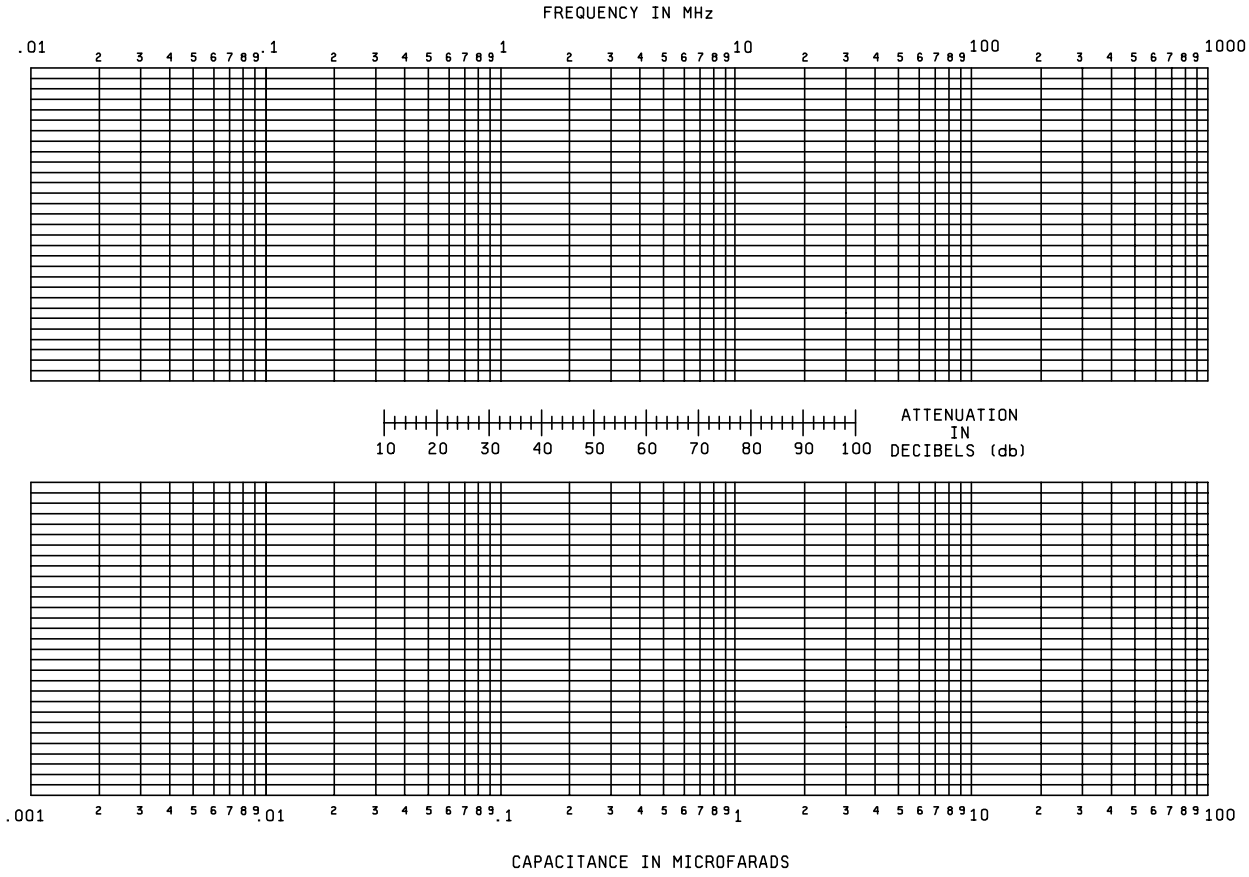



FIGURE 3. Insertion-loss chart for ideal capacitors.



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6.5 Lockwasher. External tooth-type lockwashers should be used for all applications (see 3.1), except where space limitations, appearance, or other special conditions necessitate the use of internal tooth-type or split-type lockwashers. Internal tooth-type lockwashers should not be used with binder head screws. Screws or nuts with captive lockwashers will be permitted, provided that replacement with common-type screws and lockwashers or nuts and lockwashers may be readily effected. Soldering lug and lockwasher combinations should not be used.

6.6 Definitions.

6.6.1 Radio interference. Radio interference is defined as undesired conducted or radiated electrical disturbances, including transients, within the range of frequencies covered by this specification, which may interfere with the operation of electrical or electronic communication equipment or other electronic equipment.

6.6.2 Hermetic sealing. For the purpose of this specification, a hermetically sealed capacitor is one in which the capacitance element is contained within a sealed enclosure of ceramic, glass or metal, or combinations thereof, where sealing is accomplished by material fusion, welding, brazing or soldering.

6.7 Supersession data. ER capacitors of this specification supersede similar non ER items (CZR23 for CZ23, and CZR24 for CZ24) of this specification and similar items from MIL-C-39011, (Capacitors, Feed Through, Radio-Interference Reduction, ER).

6.8 Required construction.

Extended foil: Aluminum foil electrodes separated by paper or plastic or paper-plastic dielectric and noninductively wound.

6.9 Subject term (key word) listing.

Dielectric withstanding voltage
Dissipation factor
Insulation resistance

6.10 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.



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APPENDIX

PROCEDURE FOR QUALIFICATION INSPECTION

10. SCOPE

10.1 This appendix details the procedure for submission of samples for qualification inspection of capacitors covered by this specification. The procedure for extending qualification of the required sample to other capacitors covered by this specification is also outlined herein.

20. SUBMISSION

20.1 Sample.

20.1.1 Single-style submission. A sample consisting of 36 specimens of the type for which qualification is sought shall be submitted (see 30.1).

20.1.2 Combined-type submission. A maximum of two types in each style for which qualification Is sought, consisting of 18 specimens of the highest watt-second rating or volt-ampere reactive, as applicable in the highest voltage rating and 18 specimens of the lowest capacitance value in any voltage rating, shall be submitted. Table XI is an example of a combined-type submission (see 30.2).

TABLE XI. Example of combined-type submission.

| Style | Type to be submitted | Number of samples to be submitted | Watt-second rating 1/ | Voltage, volts |
|---|---|---|---|---|
| CZ23 | CZ23BKF103 | 18 | 0.0018 | 600 DC |
| CZ23 | CZ23BKF474 | 18 | 0.0846 | 600 DC |
| CZ24 | CZ24BKF103 | 18 | 0.0018 | 600 DC |
| CZ24 | CZ24BKF474 | 18 | 0.0846 | 600 DC |
| CZR23 | CZR23BKF474M | 32 | 0.0846 | 600 DC |
| CZR24 | CZR24BKF103M | 32 | 0.0018 | 600 DC |

1/ The watt-second rating will be determined by the following formula:

$$W = \frac{CE^2}{2}$$

Where: W = energy content in watt-seconds.
C = nominal capacitance in farads.
E = dc voltage rating in volts.

30. EXTENT OF QUALIFICATION

30.1 Single-type submission. Qualification will be restricted to the type submitted.

30.2 Combined-type submission. Qualification of the types submitted will be extended to types whose watt-second and voltage ratings are not greater than the highest watt-second and voltage ratings submitted, and whose capacitance values are not less than the lowest capacitance value submitted. In addition, submission and qualification of a type or types in one characteristic automatically carries with it qualification of the lower characteristics, provided the units are of the same style, material, impregnant or filling compound and construction (e.g. qualification of characteristic K provided the basis for qualification of characteristics K and E). Qualification of ER parts will give qualification of non-ER parts of the same dielectric, construction, voltage ratings, current ratings and capacitance values.



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Preparing activity:
DLA - CC

(Project 5910-2124)



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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

2. The submitter of this form must complete block1s 4, 5, 6, and 7, and send to preparing activity.

3. The preparing activity must provide a reply within 30 days from receipt of the form.

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| **I RECOMMEND A CHANGE:** | **1. DOCUMENT NUMBER** MIL-PRF-11693D | **2. DOCUMENT DATE** (YYMMDD) 010809 |
|---|---|---|

**3. DOCUMENT TITLE** CAPACITORS, FEED THROUGH, RADIO-INTERFERENCE REDUCTION, DC (HERMETICALLY SEALED IN METAL CASES), ESTABLISHED AND NON-ESTABLISHED RELIABILITY, GENERAL SPECIFICATION FOR

**4. NATURE OF CHANGE** (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

**5. REASON FOR RECOMMENDATION**

**6. SUBMITTER**

| a. NAME (Last, First, Middle initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Incl Area Code)<br>(1) Commercial<br>(2) DSN<br>(If applicable) | 7. DATE SUBMITTED<br>(YYYYMMDD) |

**8. PREPARING ACTIVITY**

| a. NAME<br>Defense Supply Center, Columbus<br>ATTN: VAT<br>michael.radecki@dscc.dla.mil | b. TELEPHONE (Include Area Code)<br>(1) Commercial (614) 692-0561 (2) DSN 850-0561 |
|---|---|
| c. ADDRESS (Include Zip Code)<br><br>PO Box 3990<br>Columbus OH 43216-5000 | **IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:**<br>Defense Standardization Program Office (DLSC-LM)<br>8725 John J. Kingman Road, Suite 2533<br>Fort Belvoir, Virginia 22060-6221<br>Telephone (703) 767-6888 DSN 427-6888 |



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